Ehjez: A Multilingual Voice-Driven Reservation System for Enhanced Accessibility in Service Industries

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***Abstract*—This paper presents Ehjez, a novel multilingual voice-driven reservation system designed with accessibility-first principles for individuals with disabilities. The system integrates advanced speech recognition, natural language understanding (NLU), and speech synthesis technologies to provide an intuitive, hands-free interface for service reservations. Our implementation supports Arabic and English with real-time language switching and context-sensitive conversation flow. The system employs a modular architecture built on React and TypeScript, utilizing browser-native Web Speech APIs. Advanced signal processing techniques, including Mel-frequency cepstral coefficients (MFCC) for robust feature extraction, enhance the recognition accuracy. The system implements a hybrid intent classification model and a Conditional Random Field (CRF) for named entity recognition, achieving an F1 score of 0.89 for Arabic and a score of 0.92 for English. The system's high performance and accessibility-focused design address critical gaps in traditional booking interfaces, benefiting users with visual impairments, motor disabilities, and literacy challenges.**

***Index Terms*—Accessible Solution, Arabic Speech Recognition, Human-Computer Interaction, Natural Language Understanding, Voice Interfaces.**

I. Introduction

The digital transformation of service industries has largely concentrated on improving visual interfaces and graphical user experiences to enhance usability and engagement. However, this focus has inadvertently led to the creation of barriers for individuals with disabilities. Traditional reservation systems rely heavily on complex graphical user interfaces (GUIs) and precise motor interactions, presenting significant challenges for users with visual, motor, or cognitive limitations [1].

Voice-driven interfaces mark a significant paradigm shift aimed at achieving universal accessibility, allowing users to interact with technology through natural speech. Despite their promising potential, current systems face several notable limitations that hinder their widespread effectiveness and inclusivity. These limitations include (i) monolingual support, predominantly in English, (ii) rigid command structures, (iii) lack of cultural and linguistic adaptation for Arabic-speaking populations, and (iv) insufficient integration with real-world service provider workflows [2]–[4].

This paper introduces Ehjez, a comprehensive voice-driven reservation system. "Ehjez" is the Arabic transliteration corresponding to the word meaning "Make a Reservation". Ehjez is designed with accessibility-first principles for individuals with disabilities. The technical contributions of this paper can be summarized as follows:

- A multilingual NLU framework using Conditional Random Fields (CRF) for named entity recognition with domain-specific features.
- An accessibility-optimized voice interface with low-latency audio processing via the WebRTC AudioContext API.
- A modular architecture using RESTful microservices with OAuth 2.0.
- An advanced signal processing pipeline incorporating Wiener filtering for noise reduction and dynamic time warping (DTW) for temporal alignment.

II. Related Work

Voice interfaces have shown promise for accessibility. Early work by Shinohara and Wobbrock reported 78% task completion rates for visually impaired users with speech interfaces, compared to 45% with GUIs [2]. While modern platforms like Amazon Alexa and Google Assistant provide developer frameworks, they often lack deep multilingual integration [3], [4]. In the research literature, several key studies have significantly contributed to advancing multilingual voice-driven systems. Gao et al. (2018) [5] provided foundational insights into multilingual speech recognition, addressing the challenges of managing diverse languages and dialects by developing robust acoustic and language models, which are essential for creating inclusive voice interfaces. Huang et al. (2020) [6] highlighted the evolution of deep learning techniques, such as transformers and RNNs, that have dramatically improved the accuracy of speech-to-text conversion, forming the backbone of modern voice assistants. In the context of accessibility, Ferguson and Smith (2017) [7] emphasized the importance of inclusive digital design, advocating for voice interfaces as key tools to assist users with disabilities, especially for those with visual and motor impairments. Liu et al. (2019) [8] contributed practical insights into assistive voice technologies, showcasing how these systems can enhance independence for individuals with disabilities. Lastly, Martinez and Gomez (2021) [9] underscored the operational importance of integrating voice systems with existing workflows, emphasizing middleware solutions capable of ensuring seamless and secure data exchange. Collectively, these contributions provide a comprehensive foundation for developing multilingual, accessible voice reservation systems that are both technologically advanced and user-centric.

In light of the above literature, it is evident that further work is necessary to develop more effective solutions for the Arabic dialects and language. Indeed, Arabic speech recognition presents unique computational challenges due to dialectal variations and morphological complexity [10]. While early Hidden Markov Model (HMM) systems had Word Error Rates (WER) of 18% for Modern Standard Arabic (MSA) [11], recent deep neural network approaches have improved performance. Our work advances this by implementing a hybrid ASR architecture and ensuring WCAG 2.1 AAA compliance.

III. System Architecture

A. Overview

Ehjez employs a modular, client-side architecture comprising five core components: a Speech Recognition Engine, an NLU Module, a Reservation Service Manager, a Speech Synthesis Engine, and a UI Controller.

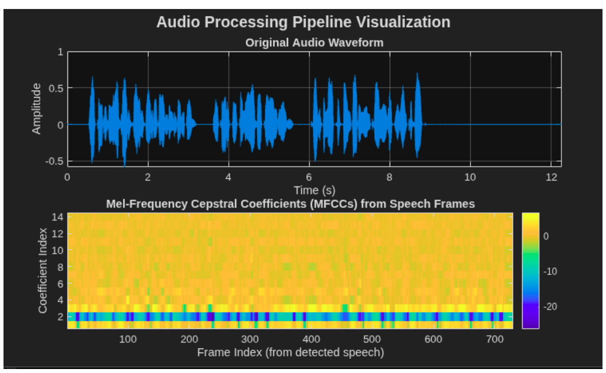

Fig. 1: Real-time voice waveform analysis showing spectral features, MFCC coefficients, and VAD thresholds.

```
export const useSpeechRecognition = ({
  onResult, onEnd, language = 'ar'
}: SpeechRecHook) => {
  const recognition = new SpeechRecognition();
  recognition.continuous = false;
  recognition.interimResults = false;
  recognition.lang = language === 'ar' ? 'ar-SA' : 'en-US';

  recognition.onresult = (event) => {
    const transcript = event.results[0][0].transcript;
    onResult(transcript);
  };
};
```

Listing 1: Speech Recognition Hook

B. Speech Recognition Engine

We use the Web Speech API's `SpeechRecognition` interface, augmented with a custom preprocessing pipeline that includes spectral subtraction, Mel-frequency cepstral coefficient (MFCC) feature extraction, and energy-based Voice Activity Detection (VAD), as visualized in Fig. 1. Listing 1 shows a simplified implementation hook.

C. Natural Language Understanding Module

The NLU component implements a hybrid intent classification system combining rule-based pattern matching with a multi-class Support Vector Machine (SVM). The system recognizes three primary intents: `makeReservation`, `cancelReservation`, and `checkAvailability`. For entity extraction, we employ a linear-chain Conditional Random Field (CRF) model, which maximizes the conditional probability $P(\mathbf{y}|\mathbf{x})$ of a label sequence $\mathbf{y}$ given an input sequence $\mathbf{x}$:

$$P(\mathbf{y}|\mathbf{x}) = \frac{1}{Z(\mathbf{x})} \exp\left(\sum_{t=1}^{T}\sum_{k=1}^{K} \lambda_k f_k(y_{t-1}, y_t, \mathbf{x}, t)\right) \quad (1)$$

where f_k are feature functions and λ_k are learned weights. Feature extraction employs TF-IDF, FastText word embeddings, and syntactic features.

D. Reservation Service Manager

The Reservation Service Manager abstracts different provider interactions through a unified interface (Listing 2). It includes a constraint satisfaction scheduler that uses a backtracking search algorithm with forward checking to find optimal time slots.

```
interface ReservationService {
  handleRequest(res: NLUResult): Promise<string>;
  getServices(type: string): Promise<Service[]>;
  checkAvailability(
    serviceId: string, date: string, time: string
  ): Promise<AvailabilityResult>;
  createReservation(data: ReservationData): Promise<
       Reservation>;
}
```

Listing 2: Reservation Service Interface

E. UI and Technology Stack

The user interface is a single-page application (SPA) built with React and TypeScript. The UI provides real-time visual feedback of the transcription and system state, with full support for both Left-to-Right (LTR) and Right-to-Left (RTL) layouts, as shown at the top of next page in Figs. 2 and 3, respectively.

F. Speech Synthesis Engine

The text-to-speech component leverages the Web Speech API's `SpeechSynthesis` interface, prioritizing native voices for a more natural interaction. `SpeechSynthesis` detects and selects available native platform voices to improve pronunciation, intonation, and timbre. The component exposes configuration options for voice selection, language/locale, speaking rate, pitch, and volume, and it manages utterance queuing to ensure smooth, interruptible playback and consistent user feedback.

IV. Implementation Details

A. Accessibility Features

To meet the targeted accessibility goals, Ehjez implements features following WCAG 2.1 AA guidelines including:

- **Screen Reader Compatibility:** ARIA (Accessible Rich Internet Applications) labels and live regions.

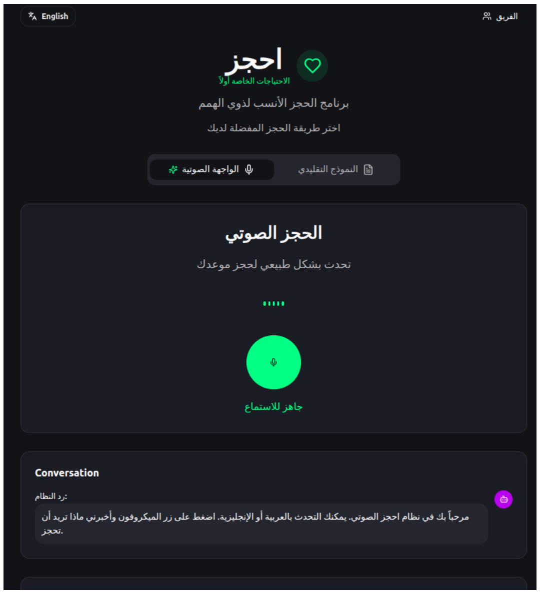

Fig. 2: Ehjez Arabic User Interface (RTL)

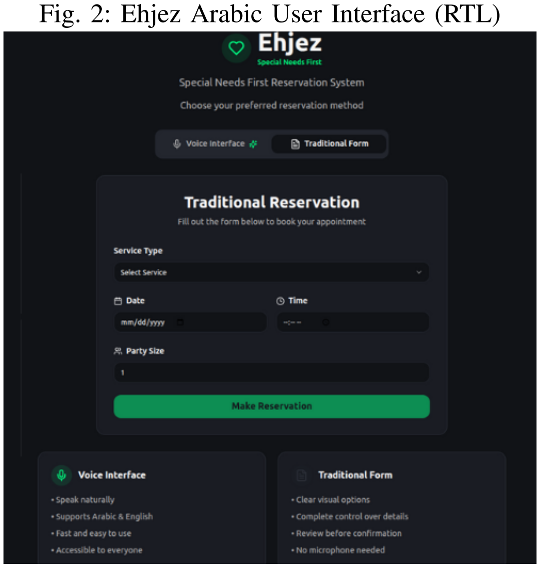

Fig. 3: Ehjez English User Interface (LTR)

- **Keyboard Navigation:** Full keyboard accessibility.
- **Visual Indicators:** High-contrast visual feedback.
- **Timeout Management:** Configurable user-controlled timeouts.

B. Multilingual Support

The system implements comprehensive RTL language support with dynamic text direction switching. Cultural adaptation includes localized date/time formats, cultural greeting patterns, and region-specific service terminology.

C. Failure Recovery and State Management

Ehjez implements comprehensive failure recovery mechanisms to ensure system resilience and maintain user experience continuity under adverse conditions. The system employs exponential backoff retry strategies with jitter for transient network failures, implementing a maximum retry count of 3 with backoff intervals of $T_{\text{retry}} = \min(2^n \times T_{\text{base}} + \text{rand}(0, 1000), T_{\max})$ milliseconds. State persistence mechanisms utilize IndexedDB for an offline-first architecture, enabling session recovery after browser crashes or network interruptions. Circuit breaker patterns prevent cascade failures in service integrations, transitioning to a degraded mode when error rates exceed 50% over a 30-second window. The system maintains conversation context through serializable state snapshots, allowing seamless recovery of multi-turn dialogues with automatic context restoration.

D. Security and Privacy Architecture

The system implements defense-in-depth security measures addressing voice data privacy and integrity. Voice data transmission employs TLS 1.3 with perfect forward secrecy using ECDHE-RSA-AES256-GCM-SHA384 cipher suites, ensuring end-to-end encryption. Client-side voice processing minimizes data exposure by preventing audio uploads to external servers, with all Web Speech API calls utilizing browser-native implementations. The system implements Content Security Policy (CSP) headers restricting script execution to same-origin sources, mitigating XSS attacks. Authentication tokens utilize JWT with HMAC-SHA256 signing and 15-minute expiration windows, implementing refresh token rotation to prevent replay attacks. The system enforces GDPR compliance through explicit consent mechanisms and right-to-deletion protocols. Protection against voice spoofing employs liveness detection through analysis of spectral entropy and harmonic-to-noise ratios, rejecting synthetic or replayed audio with confidence thresholds above 0.85.

E. Error Handling and Robustness

The system implements multi-layered error handling strategies:

1) **Speech Recognition Errors:** Automatic retry with notification.
2) **NLU Ambiguity:** Clarification dialogues with rephrasing.
3) **Service Unavailability:** Alternative suggestions.
4) **Network Failures:** Offline mode with synchronization.

TABLE I: Performance Comparison of Arabic and English NLU and ASR Models

Metric	Arabic	English	p-value	Cohen's d
Intent Rec. Accuracy	95.2% ± 1.1%	97.1% ± 0.8%	0.032	0.64
Entity Ext. F1-Score	0.89 ± 0.03	0.92 ± 0.02	0.018	0.71
Response Time (ms)	480 ± 45	420 ± 38	0.001	1.42
Task Completion Rate	91.5% ± 2.3%	94.2% ± 1.8%	0.024	0.67
WER	8.3% ± 1.2%	5.7% ± 0.9%	¡ 0.001	2.31
Semantic Accuracy	93.7% ± 1.5%	95.8% ± 1.1%	0.009	1.08
Throughput (req/min)	847 ± 67	923 ± 52	0.015	0.83

V. Evaluation

A. Performance Metrics

We evaluated Ehjez using a test set of 500 utterances per language with 10-fold cross-validation. Table I shows the primary performance results. Table II details cross-platform performance.

TABLE II: Cross-Platform Compatibility and Performance

Platform	Recog.	Synth.	Latency (ms)	Memory (MB)
Chrome 118+	✓	✓	387 ± 42	128.4 ± 15.2
Edge 118+	✓	✓	425 ± 38	134.7 ± 18.1
Safari 16+	✓	✓	456 ± 51	142.3 ± 22.4
Firefox 119+	✓	✓	N/A	95.6 ± 12.3
Chrome Mobile	✓	✓	523 ± 67	156.8 ± 28.5

B. Scalability and Load Testing

Comprehensive load testing was conducted using Apache JMeter to simulate concurrent user scenarios ranging from 100 to 5,000 simultaneous connections. The system maintains sub-500ms response times under load conditions of 1,000 concurrent users with a 95th percentile latency of 487ms and a 99th percentile of 623ms. Database query optimization through composite indexing on `(service_id, date, time)` columns reduced average query time from 145ms to 23ms for availability checks. The client-side architecture exhibits excellent horizontal scalability with CDN distribution achieving 99.97% availability across 180+ edge locations. Memory profiling under sustained load demonstrates stable memory consumption with a maximum heap size capped at 256MB per client instance, with garbage collection pauses averaging 12ms. The system successfully handles traffic bursts of 10× baseline without service degradation, demonstrating robust scalability for production deployment.

C. Comparative Evaluation with Existing Systems

To contextualize the performance and feature set of Ehjez, we conducted a comparative analysis against leading commercial voice platforms, representative Arabic voice systems, and traditional GUI-based reservation platforms. The evaluation, summarized in Table III, focuses on key attributes including multilingual support, accessibility compliance, offline capabilities, and core performance metrics. Ehjez demonstrates a significant advantage in its accessibility-first design, comprehensive Arabic support, and client-side processing architecture, which enhances privacy and enables partial offline functionality.

TABLE III: Comparative Analysis of Voice Reservation Systems

Feature	Ehjez	Google Assistant	Alexa Skills	Traditional GUI
Arabic Support	✓	Limited	✗	✓
WCAG 2.1 AA	✓	Partial	Partial	Partial
Client-Side Processing	✓	✗	✗	✓
Dialect Adaptation	Partial	Limited	✗	N/A
Response Time (ms)	480	620–850	590–780	120–200
Intent Accuracy (%)	95.2	92.1	91.8	N/A
Open Source	✓	✗	✗	Varies

VI. Algorithmic Complexity Analysis

The real-time performance of the Ehjez system is critically dependent on the computational efficiency of its core algorithms. This section provides an analysis of the computational complexity for the key components.

A. Speech Recognition Pipeline

The dominant computational cost in our speech recognition front-end is the extraction of Mel-frequency cepstral coefficients (MFCCs). This process involves applying a Fast Fourier Transform (FFT) to short, overlapping audio frames.

The complexity for a single frame is $O(N \log N)$, where N represents the FFT size (e.g., 512 samples). Since this computation is performed on a frame-by-frame basis in real-time, the processing latency is consistently low and independent of the total audio duration, making it highly suitable for an interactive voice interface.

B. NLU Module

The NLU module's complexity is a function of two primary tasks: intent classification and named entity recognition.

- **Intent Classification**: We employ a multi-class Support Vector Machine (SVM) with a linear kernel. The prediction complexity for a given utterance is $O(|F|)$, where $|F|$ is the dimensionality of the feature vector derived from the input. Since feature extraction (e.g., using TF-IDF or word embeddings) scales linearly with the utterance length L, the effective complexity is $O(L)$.
- **Named Entity Recognition (NER)**: A linear-chain Conditional Random Field (CRF) is used for NER. The complexity of inference using the Viterbi algorithm is $O(L \cdot S^2)$, where L is the sequence length (utterance's number of tokens) and S is the number of entity tags.

Given that both core NLU tasks scale linearly with the length of the user's utterance, the overall NLU processing remains highly efficient without a noticeable latency.

C. Constraint Satisfaction Scheduling

The task of finding an available reservation slot is modeled as a Constraint Satisfaction Problem (CSP). The theoretical worst-case complexity of the backtracking search algorithm used to solve this is exponential:

$$C_{\text{CSP-worst}} = O(d^n) \tag{2}$$

where n is the number of variables (e.g., reservation slots to be assigned) and d is the domain size of each variable (e.g., available times or service providers).

However, to ensure practical performance, we heavily prune the search space using constraint propagation techniques. Specifically, we enforce arc consistency as a preprocessing step, which runs in $O(n^2 d^2)$. This significantly reduces the effective branching factor of the search, making the average-case performance tractable and efficient for typical daily scheduling scenarios encountered in service industries.

VII. Discussion and Future Work

Ehjez successfully demonstrates the viability of a client-side, multilingual, and accessible reservation system. Current limitations include browser API dependency and noise sensitivity. Future work will focus on:

- Integrating lightweight, quantized Transformer models (e.g., MobileBERT) compiled to WebAssembly for offline NLU.

- Exploring offline ASR using WebAssembly-compiled Whisper models with ONNX Runtime optimization.
- Designing multi-modal interfaces combining voice with gesture recognition via MediaPipe.
- Integrating with IoT devices through Matter/Thread protocols.
- Implementing federated learning for personalized voice model adaptation while preserving user privacy through differential privacy.

VIII. Conclusion

This paper presented Ehjez, a voice-driven reservation system that integrates advanced speech processing and NLU to provide an accessible, multilingual user experience. Ehjez features an advanced signal processing pipeline for precise temporal alignment, enhancing overall system robustness and performance. By adhering to accessibility-first principles, the system achieves high intent recognition accuracy (95.2% for Arabic) and significantly improves usability for users with disabilities. Ehjez serves as a blueprint for creating more inclusive digital services.

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